

TrES-3b

R-2006

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_130504 · created 2026-08-07T16:39:36+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2456416.8784 +/- 0.0011 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.007 +/- 0.056
Transit depth (Rp/Rs)^2	0.0055 +/- 0.083 [%]
Orbital Inclination (inc)	78.8 +/- 2.86
Airmass coefficient 1 (a1)	1.63 +/- 0.17
Airmass coefficient 2 (a2)	-0.46 +/- 0.79
Scatter in the residuals of the lightcurve fit is	9.92 %
Best Comparison Star	#2 - [532, 141]
Optimal Aperture	4.24
Optimal Annulus	9.72
Transit Duration (day)	0.011 +/- 0.019

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.007 ± 0.056 vs 0.16309 (deviation 2.79σ)
Duration fit vs published	0.011 d vs 0.0567 d (deviation 2.4σ)
Mid-time O–C	0.9 min
Depth SNR	0.1
Residual scatter	9.92 %

Light curve

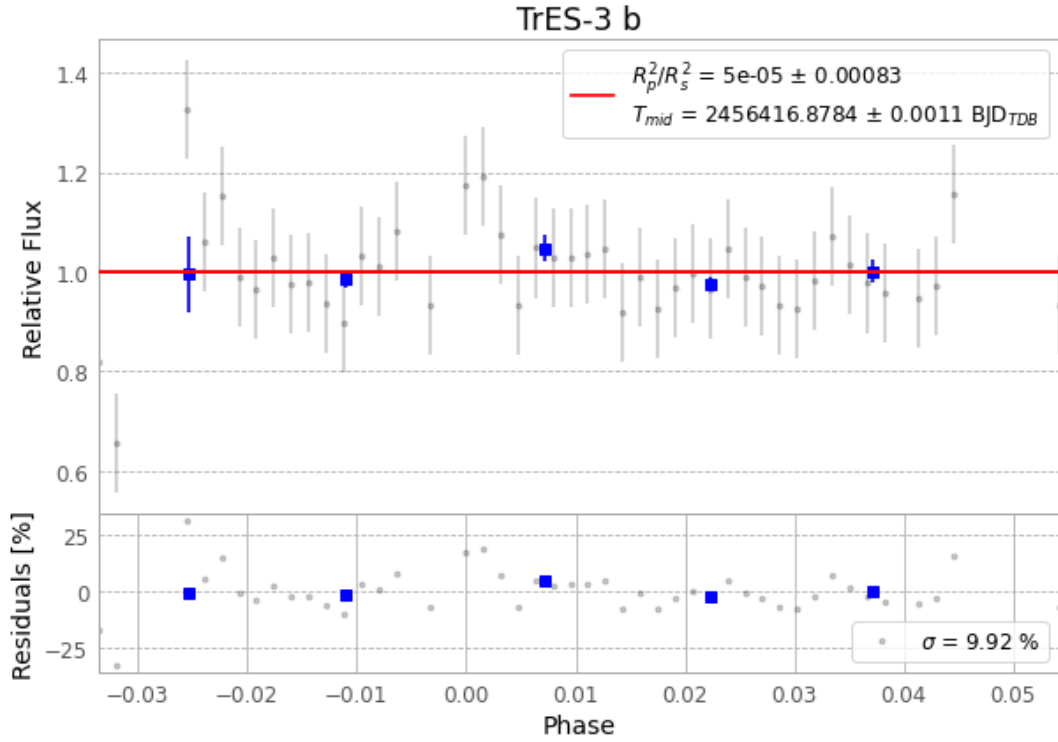


Figure 1 — FinalLightCurve_TrES-3 b_2013-05-04.png (SHA-256 68e6745e2d34bb08...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [326, 250], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 0ecd190b411f4c81...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit fd74a91

Data provenance

56 FITS frames (37 MB), TRES-3130504075712.FITS → TRES-3130504104212.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-130504.log (02f8a6d96b90...), FinalLightCurve_TrES-3 b_2013-05-04.png (68e6745e2d34...), FinalLightCurve_TrES-3 b_2013-05-04.csv (ce8a2c7f63ae...)

Compute resources

Wall time 174.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-07 16:39Z.